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SPECIAL ISSUE ON INTRASPECIFIC VARIATION IN PLANT FUNCTIONAL TRAITS

Intraspecific trait variation and reversals of trait strategies across key climate gradients in native Hawaiian plants and non-native invaders

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• **Background and Aims** Displacement of native plant species by non-native invaders may result from differences in their carbon economy, yet little is known regarding how variation in leaf traits influences native–invader dynamics across climate gradients. In Hawaii, one of the most heavily invaded biodiversity hotspots in the world, strong spatial variation in climate results from the complex topography, which underlies variation in traits that probably drives shifts in species interactions.
• **Methods** Using one of the most comprehensive trait data sets for Hawaii to date (91 species and four islands), we determined the extent and sources of variation (climate, species and species origin) in leaf traits, and used mixed models to examine differences between natives and non-native invasives.
• **Key Results** We detected significant differences in trait means, such that invasives were more resource acquisitive than natives over most of the climate gradients. However, we also detected trait convergence and a rank reversal (natives more resource acquisitive than invasives) in a sub-set of conditions. There was significant intraspecific trait variation (ITV) in leaf traits of natives and invasives, although invasives expressed significantly greater ITV than natives in water loss and photosynthesis. Species accounted for more trait variation than did climate for invasives, while the reverse was true for natives. Incorporating this climate-driven trait variation significantly improved the fit of models that compared natives and invasives. Lastly, in invasives, ITV was most strongly explained by spatial heterogeneity in moisture, whereas solar energy explains more ITV in natives.
• **Conclusions** Our results indicate that trait expression and ITV vary significantly between natives and invasives, and that this is mediated by climate. These findings suggest that although natives and invasives are functionally similar at the regional scale, invader success at local scales is contingent on climate.

Key words: Abiotic gradients, ecophysiology, environmental filtering, functional trait, Hawaii, intraspecific trait variation, invasion, island, leaf economy, trait convergence.

INTRODUCTION

Biological invasions threaten native species biodiversity on a global scale (Simberloff, 2000; Vilà *et al*., 2011; Pysek *et al*., 2012), and yet the mechanisms that facilitate invasion remain poorly understood (Levine *et al*., 2003; Pauchard and Shea, 2006; Melbourne *et al*., 2007; Moles *et al*., 2012). Differences in physiological and morphological traits associated with carbon economy [i.e. the leaf economic spectrum (LES)] are thought to promote invasiveness in plants, such that invasive species can acquire and utilize resources faster than natives (Leishman *et al*., 2007; Ordonez *et al*., 2010; Peñuelas *et al*., 2010*b*). An acquisitive resource use strategy among invasives, coupled with fast growth and high reproductive output, is thought to drive the displacement of native species throughout their geographic ranges (Richardson and Pyšek, 2006; Moles *et al*., 2012). However, within any given species' geographic range, traits often vary in response to resource gradients such as rainfall, temperature, soil nutrients and light availability (Lavorel and Garnier, 2002; Reich and Oleksyn, 2004; Wright *et al*., 2005; Ordoñez *et al*., 2009; Atkin *et al*., 2015). As a result of this phenotypic variation, native species may be outperformed by invasive species in some environments but not in others (Hulme and Bernard-Verdier, 2018). This scenario would result in a variable landscape of invader success that is climate dependent.

To better understand the performance of native and invasive species over climate gradients requires studies that elucidate drivers of trait variation and clarify the environmental conditions under which natives are likely to be outperformed by invasive species (Hulme and Bernard-Verdier, 2018). The importance of within-species trait variation [hereafter, intraspecific trait variation (ITV)] to ecological and evolutionary processes is well accepted (Albert *et al*., 2010; Bolnick *et al*., 2011; Violle *et al*., 2012; Siefert *et al*., 2015), yet few studies have compared patterns of ITV for multiple species across environmental gradients (but see Lepš *et al*., 2011; Hulshof *et al*., 2013; Bucher *et al*., 2016). In the context of biological invasions, research focused on comparing mean species traits

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across native and non-native species argues that variation between species is larger or more important than ITV (Kueffer *et al*., 2008; Kattge *et al*., 2011; Godoy *et al*., 2012). However, explicitly incorporating species-specific ITV across environmental gradients might facilitate our ability to detect differences in traits across native and non-native species and better understand the mechanisms that facilitate invasion.

Climate plays an important role in shaping plant traits at multiple spatial scales. Regional and global studies have identified mean annual rainfall as a major source of trait variation, such that under increasingly arid conditions plants tend to have leathery dense leaves [high leaf mass per area (LMA)] with greater nitrogen content per unit area (Wright *et al*., 2005; Cernusak *et al*., 2011). Other climate variables also contribute to ITV such that greater mean annual temperatures generally result in longer leaf life span and greater LMA in deciduous species, while greater levels of irradiance are associated with longer-lived leaves and lower photosynthetic rates for a given leaf nitrogen content (Wright *et al*., 2005). Landscape-scale patterns in ITV thus reflect spatial heterogeneity in climate, resulting in variation among plant communities that may arise from species sorting by climate or variation within species due to population differentiation or phenotypic plasticity. At more local scales, trait variation within plant communities may result from niche differentiation or microscale habitat variation (Messier *et al*., 2017).

An ability to rapidly acquire and utilize resources such as light, water and soil nutrients may promote dominance of invasives over resource-conservative native species when resource availability is high (Richards *et al*., 2006), while resource-conservative natives are more likely to have an advantage in resource-limited environments. A review by Daehler (2003) noted that when resources are limited, native species perform equally well if not better than invasive species. In contrast, invasive species commonly thrive in disturbed habitats, characterized by high resource availability (Davis *et al*., 2000). In cases where invaders establish in the absence of such disturbances, more information is needed to understand the mechanisms facilitating invasion. For example, the extent to which phenotypic plasticity enhances fitness of invasives as they move from resource-limited to resource-rich environments remains unclear (Richards *et al*., 2006; Pyšek and Richardson, 2008). Many studies have attempted to identify traits conferring invasiveness, yet relatively few of these have been conducted along such sharp gradients in resource availability, such as those that occur within the Hawaiian archipelago. In Hawaii and other oceanic islands, non-native species often become naturalized, facilitating the extinction of native plants (Sax and Gaines, 2008). The extent to which variation in climate alters resource availability and thus influences the success of invaders in Hawaii is the subject of the current study. A recent relevant study by Henn *et al*. (2019) found that native Hawaiian species and non-natives expressed trait convergence in cool, wet climates and divergence in hot, dry climates. This study suggests the potential for a role of ITV in mediating native–invader dynamics. Towards this end, we conducted a field study with the following three primary objectives. (1) The first was to describe sources and magnitudes of variation in leaf traits of native Hawaiian plants and non-native invaders. We include traits associated with the leaf economic spectrum (e.g. LMA, A_{sat}, N% and P%) as well as other traits associated with resource capture and drought tolerance (e.g. leaf thickness, leaf area and chlorophyll content). Specifically, we ask: to what extent do species origin (native or non-native invasive), species identity and climate influence variation in leaf traits? (2) We wanted to determine whether trait differences between natives and invasives become more apparent when we account for climate-driven trait variation among species. (3) Lastly, we wished to determine whether natives and invasives express convergence in traits under particular climate conditions (species origin and climate interactively affect ITV). For example, in environments with low rainfall and high temperature, is there evidence of trait convergence, such that differences between natives and invasives are negligible?

Our study encompasses large gradients in elevation and mean annual rainfall across multiple islands and for a larger sample of species than has previously been studied in Hawaii. In a companion study, we detected a significant extent of overlap in the traits of native and invasive species (A. C. Westerband, unpubl. data) at the archipelago scale. A high degree of trait overlap should confer some resistance of native species to invasion due to niche saturation (Funk *et al*., 2008), and yet the very high rates of invasive plant cover and native plant declines in Hawaii suggest weak invasion resistance, which Funk *et al*. (2008) suggest results from the plasticity of invasives. We hypothesize that ITV is extensive, and that climate underlies a significant portion of this ITV, leading to faster carbon economics (and thus greater performance) of invasives compared with natives in some climates but not others. This variable performance would explain why some studies on single islands, where environmental conditions are distinct from those on other islands, detect significant trait differences while others do not (Baruch and Goldstein, 1999; Stratton and Goldstein, 2001; Asner *et al*., 2006; Funk and Vitousek, 2007; Ostertag *et al*., 2009; Peñuelas *et al*., 2010*a*). Native–invader comparisons are likely to be scale dependent (Diez *et al*., 2008; Gross *et al*., 2013; Hulme and Bernard-Verdier, 2018) and thus require studies such as this that utilize multiscale approaches.

MATERIALS AND METHODS

Study system

In Hawaii, climate is highly variable spatially, due to extreme topography resulting from the volcanic origin. Elevational gradients may thus represent a good proxy for climate variability, with considerable reductions in temperature and significant but region-specific shifts in rainfall as elevation increases (Körner, 2007). The Hawaiian Islands also vary in age, such that islands with similar climates may still express significant variation in soil nutrient concentrations (Crews *et al*., 1995; Riley and Vitousek, 1995). To include as much of this abiotic variability as possible, we sampled leaf traits across elevational gradients at 45 study sites distributed among the four largest Hawaiian Islands: Hawaii Island, Maui, Oahu and Kauai, between August 2016 and November 2017. Of the samples studied, 67 % individuals were within evergreen forest, 21 % fell within developed areas (including roadsides) and 9 % fell within shrub/scrublands

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(National and Cover Database, http://www.databasin.org/). We sampled plants across large gradients in elevation (sea level to 2000 m asl), mean annual precipitation (400–5600 mm) and mean annual air temperature (10–24 °C). Several species were sampled at multiple sites, while rare endemics may have only occurred at one or two sites (Supplementary data Table S1).

A total of 91 plant species were sampled, which includes 60 native and 31 non-native species (Supplementary data Table S2). We hereafter refer to non-natives as invasives because 26 of the 31 non-natives are considered invasive species, here defined as non-natives with high rates of population growth and spread, which pose a threat to native species abundance and diversity, and alter ecosystem function. Native status and invasiveness were determined by cross-referencing online floras (Smithsonian Institution, https://naturalhistory2.si.edu/botany/hawaiianflora/) and risk assessments (Daehler *et al*., 2004) (Pacific Islands Ecosystems at Risk, PIER, http://www.hear.org/pier/). Initially, we focused our sampling efforts on species that co-occur with the dominant woody tree species in Hawaii, *Metrosideros polymorpha*, based on a community assembly study conducted across the Hawaiian Islands by K. Barton and T. Knight (Craven *et al*., 2018). We later expanded the study to include sites dominated by other common native species such as *Acacia koa*. It was not possible to sample all species within these sites, therefore we included dominant as well as rare species to more accurately represent the community. Rare species were occasionally represented by a single individual per site (Supplementary data Table S1). It was not possible to calculate the extent of intraspecific variation for these species and they were omitted from analyses that tested relationships between climate and ITV. In total we sampled 874 individuals, comprising 560 native and 314 invasive plants (Supplementary data Table S2).

Leaf traits

For each individual, we measured ten functional traits, i.e. morphological, physiological or phenological traits that are thought to influence whole-plant performance or fitness: LMA (g m^{-2}), leaf thickness (mm), leaf area (cm^2), chlorophyll content (arbitrary units), photosynthetic assimilation rate (μmol CO_2 m^{-2} s^{-1}), transpiration (mmol H_2O m^{-2} s^{-1}), water use efficiency (WUE, μmol CO_2 $mmol^{-1}$ H_2O), leaf nitrogen (%, N_{mass}), carbon:nitrogen ratio (C_{mass}/N_{mass}) and leaf phosphorus (%, P_{mass}). The physiological variables were area based (e.g. A_{area} rather than A_{mass}), while LMA, nitrogen and phosphorus were all mass-based estimates. All traits were measured in the field with the exception of LMA, leaf area and leaf nutrient content, and physiological measurements were conducted on the same leaf.

Photosynthesis and transpiration were measured using a LI-6400/XT portable photosynthesis system (LI-COR, Lincoln, NE, USA) on the youngest fully mature leaf at a light (photosynthetically active radiation, PAR) level of 1500 μmol m^{-2} s^{-1}. For all measurements, ambient CO_2 concentration was set to 400 μmol mol^{-1} and relative humidity was maintained between 50 and 80 %. Leaf temperature was adjusted to stay below 30 °C. Mean leaf temperature was 25 ± 4 °C (s.d.), and mean relative humidity was 62 ±12 %. We used a SPAD 502 Plus Chlorophyll Meter (Spectrum Technologies, Inc, Aurora, IL, USA) to measure leaf chlorophyll content (indicated by degree of greenness) in two leaves per individual, and these measurements were averaged prior to statistical analyses.

We measured leaf thickness and leaf area on one leaf per plant. Leaf thickness was measured using a digital micrometer (Fowler High Precision, Newton, MA, USA). We then took a digital photo and estimated leaf area using ImageJ (U.S. National Institutes of Health, Bethesda, MD, USA). This leaf was oven-dried for at least 7 d at 60 °C and weighed (g) using an analytical balance. Leaf mass was divided by area to estimate LMA. We measured leaf nutrient availability by haphazardly collecting mature non-senescent leaves. For tall trees, we were only able to access low branches, while leaves on smaller trees and shrubs were collected from multiple branches and at various heights. These leaves were similarly oven-dried, and dried leaves (excluding petioles) were then finely ground before elemental analyses were carried out at the University of Hawaii at Hilo Analytical Laboratory. Because only a small amount of leaf tissue is required for analysis, species with large leaves may have only one leaf in the tissue sample (the same leaf used for LMA), while species with very small leaves had several leaves in the sample.

Climate data

For each individual plant, we recorded its geographic co-ordinates and extracted the corresponding climatic data from the Hawaii Rainfall Atlas model (Giambelluca *et al*., 2013). We selected five key abiotic variables that are commonly found to influence plant traits and performance (Pakeman *et al*., 2009; Moles *et al*., 2014; Maire *et al*., 2015). These include annual mean, annual maximum, annual minimum and the s.d. of the following variables: rainfall (mm), air temperature (°C), Penman–Monteith potential evapotranspiration (inches), relative humidity (%) and available soil moisture (ratio between 0 and 1). Potential evapotranspiration was calculated using air temperature and humidity, and is therefore correlated with both variables (see http://solar.geography.hawaii.edu/methods.html for a full description of methodology). Rainfall and soil moisture were available as monthly estimates, while daily estimates were available for the remaining variables. Elevation (m asl) was estimated using open access elevation data from Google (The Elevation API), and the elevation function in the 'elevatr' R package.

Statistical analysis

We used a principal components analysis (PCA) to reduce the climate variables to two principal component axes, which together explained 75.4 % of the variation in the abiotic data (Supplementary data Fig. S1). The first axis was most strongly correlated with soil moisture and humidity (Supplementary data Table S3), while the second axis was most strongly correlated with air temperature and potential evapotranspiration (Supplementary data Table S3). Elevation was strongly associated with the second axis as well. We thus refer to the first

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two principal component axes as the moisture axis and solar energy axis, respectively. We use the term solar energy in reference to 'available solar energy' or 'solar energy metrics', which are associated with temperature, evapotranspiration and radiation (Evans *et al*., 2005). The PCA was conducted using the 'prcomp' function. While many of the climatic variables were likely to be correlated, we included all of them in order to understand the nature of their associations.

We assessed the relative importance of four regressors: species, species origin, solar energy and moisture, on each plant trait (Objective 1) by decomposing the contribution of each regressor to the model R^2, using the 'lmg' function in the relaimpo package in R. The 'lmg' function uses sequential sum of squares from regression models with various regressor orderings that are generated using permutations. The orderings are subsequently averaged together (an unweighted average) to produce the decomposition in R^2. Relative importance values were normalized to sum to 100 %. To investigate whether ITV differs between natives and invasives, we repeated the same analyses for natives and invasives separately, using species, solar energy and moisture as the independent variables.

Generalized linear mixed models were used to determine whether traits significantly differed between native and invasive species, after accounting for species-level trait variation within a given climate (a mixed model that includes a nested structure). The model structure was as follows: Trait ~ Species Origin + Moisture + Solar Energy + (1|Species) + (1|Moisture/Species) + (1|Solar Energy/Species). In this global model, there is random variation among species independently of climate, but species also vary within a given climate level. This global model was compared to a simpler model that excluded the last two terms, i.e. including the random effect of species alone, to determine whether we would be able to more readily detect differences between natives and invasives after accounting for species variation within climate using the nested model. The nested (global) model and the non-nested model were compared using the anova function in the 'stats' R package (Objective 2). Subsequently, we used mixed models to determine whether there was a significant interaction between species origin and climate when we controlled for differences among species at a given climate (Objective 3): Trait ~ Species Origin + Species Origin:Moisture + Species Origin:Solar Energy + (1|Moisture/Species) + (1|Solar Energy/Species).

To quantify the extent to which plant traits varied within species, we estimated site-level ITV for each trait as the coefficient of variation for one species at a given site ($CV_{trait, site}$), averaged across all species ($\overline{CV}_{trait, all species}$) to produce one value per site (α term in Supplementary data Table S4). Natives and invasives were analysed separately, and species with only one replicate per site were omitted from this analysis. The response variable, site-level variation in traits, was used as a metric of ITV and regressed onto the climate variables of interest to understand how climate influences trait variation among natives and invasives (Objective 3).

For all mixed models, we evaluated the significance of the fixed effects using type II Wald χ^2 tests. We visualized differences between natives and invasives in terms of the link between climate and traits for models where the species origin by climate interaction was statistically significant. Linear mixed models were constructed using the lmer function in the 'lme4' R package, and Wald tests were conducted using the Anova function in the 'car' R package. All statistical analyses were conducted in R v. 3.5.3 and R Studio v. 1.1.423 (R Development Core Team, 2017).

RESULTS

Sources of trait variation differed widely across the leaf traits (Fig. 1), and climate (moisture and solar energy) was the greatest explanatory factor (contributing 47.5 % of variation), with species origin (native or invasive) also explaining a considerable amount at 43.8 % of the variation in traits, on average. In contrast, very little variation occurred at the species level (8.6 %). In native species, leaf traits varied most strongly with moisture availability (51.0 % of explained variation) and least strongly across species (18.1 %) (Fig. 1), while for invasives, traits varied most strongly across species (59.5 % of explained variation) and least strongly with solar energy (16.7 %) (Fig. 1). Natives and invasives expressed similar amounts of ITV (both 23%) when averaged across all traits (Supplementary data Fig. S2). Overall, both natives and invasives expressed the highest ITV in their physiological traits (photosynthesis, transpiration and WUE), and the lowest ITV in leaf chlorophyll content and leaf nutrient availability (leaf N, leaf carbon:nitrogen and leaf P). Native species expressed the highest ITV in transpiration rates, while invasives expressed the highest ITV in photosynthetic rates.

Controlling for climate-driven variation among species (species nested within climate as random effects) provided a better model fit to the data than did models that only controlled for species-level variation (species as a random effect) (Supplementary data Table S5), and this was consistent across all traits. The non-nested model detected significant trait differences between natives and invasives for the same traits as the nested model; however, the nested model was strongly preferred, i.e. it captured more variation in the data and thus provided a better fit. When we evaluated the fixed effects of models that included climate-driven variation, we detected significant differences between natives and invasives in leaf area, leaf thickness, leaf nutrient availability and chlorophyll content, whereas the physiological traits did not vary with species origin (Supplementary data Table S6).

Natives and invasives exhibited variation in their traits with climate, and there was evidence of trait convergence, which occurred at different points along the climate gradient for each trait (Figs 2 and 3; Table 1; Supplementary data Table S7). A_{sat}, N_{mass} and P_{mass} of natives and invasives converged at moderate levels of temperature and potential evapotranspiration (solar energy), whereas leaf C:N of natives and invasives converged at high levels of solar energy, and LMA and leaf thickness never converged along the solar energy gradient. With regards to moisture availability, A_{sat} and transpiration converged at high moisture, whereas leaf C:N and leaf N converged at low moisture, and WUE and P_{mass} converged at intermediate moisture. LMA, leaf thickness and leaf area never converged in response to moisture. We summarized these patterns of divergence or convergence in Table 1. Interestingly, the identity of the more resource-acquisitive group (natives or invasives) changed along the climate gradients in some traits (Table 1). For example, in cool and

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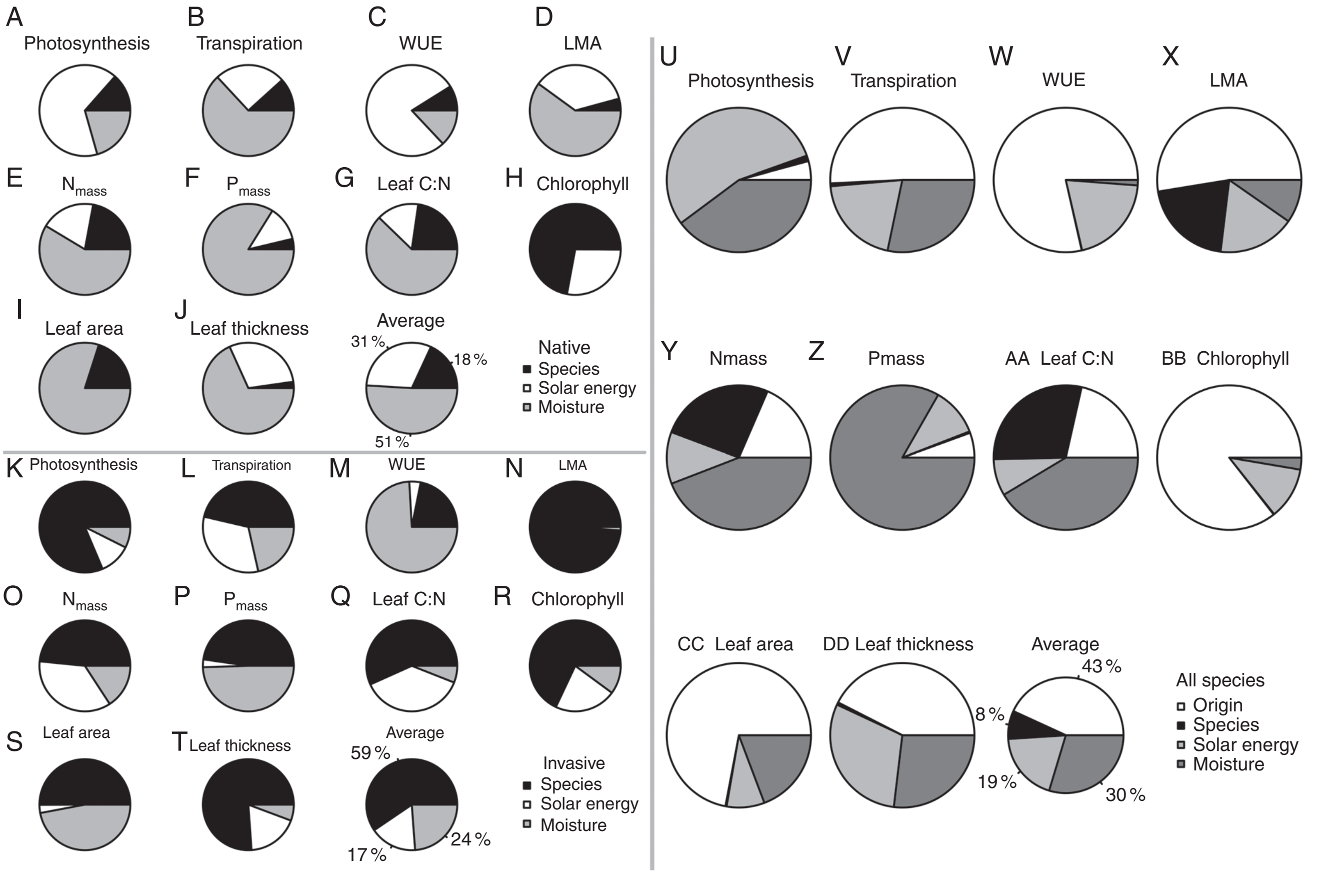

FIG. 1. Decomposition of R^2, i.e. relative importance, of plant traits in native and non-native invasive species in Hawaii. The traits include photosynthesis, transpiration, WUE, LMA, N_{mass}, P_{mass}, carbon:nitrogen, chlorophyll content, leaf area and leaf thickness. R^2 was decomposed to show the respective contributions of species, species origin (native vs. invasive), and two principal component axes representing climate: available solar energy and moisture. The final sub-plot shows the relative importance averaged across all ten traits.

less arid environments, invasives had higher A_{sat}, leaf N and leaf P (and therefore were more resource acquisitive) than natives, but the reverse was true in hotter and more arid environments. We also saw a rank reversal in WUE and P_{mass}; invasives were more water use efficient (less resource acquisitive) than natives in high moisture environments but less water use efficient under low moisture, and natives had higher P_{mass} (were more resource acquisitive) than invasives under low moisture.

Lastly, the extent of intraspecific variation in leaf traits was significantly influenced by climate in both natives and invasives, although the patterns of response varied between natives and invasives (Fig. 4; Supplementary data Table S8). In natives, there was less ITV in leaf chlorophyll content in drier locations, while there was less ITV in leaf P in hotter environments. In invasives, there was less ITV in WUE in dry environments, there was less ITV in leaf thickness in hot environments and more in wet environments and there was less ITV in transpiration and LMA in less humid environments. Overall, there were more significant associations between ITV and the climate variables in invasives than in natives (Supplementary data Table S8). ITV in leaf thickness of invasives was influenced by more climate variables than any other plant trait, suggesting strong climate-driven variation in leaf thickness (due to genetic variation or phenotypic plasticity)

DISCUSSION

Identifying traits linked to invader success remains a principal goal in ecology. Due to the well-documented link between climate and functional traits, it is likely that climate plays a key role in native–invasive trait differences. Yet, despite considerable efforts to characterize the extent to which climate influences trait means and variation both across and within species (Wright *et al.*, 2005; Kichenin *et al.*, 2013; Maire *et al.*, 2015; Freschet *et al.*, 2017; Bruelheide *et al.*, 2018), few of these studies have clarified the extent to which ITV might influence native–invader dynamics. Using the highly invaded Hawaiian Islands as a model system, we detected significant shifts in plant trait means but also trait variation in response to climate, such that invasive species shifted from being more resource acquisitive to less resource acquisitive than native species under a sub-set of environmental conditions. Additionally, ITV generally decreased with increasing aridity for both natives and non-natives. Models that explicitly included this climate-driven ITV better captured variation in the focal traits, suggesting that future comparisons of natives and invasives should carefully consider the value of using species means in regions where strong resource gradients are likely to lead to considerable ITV (Hulme and Bernard-Verdier, 2018).

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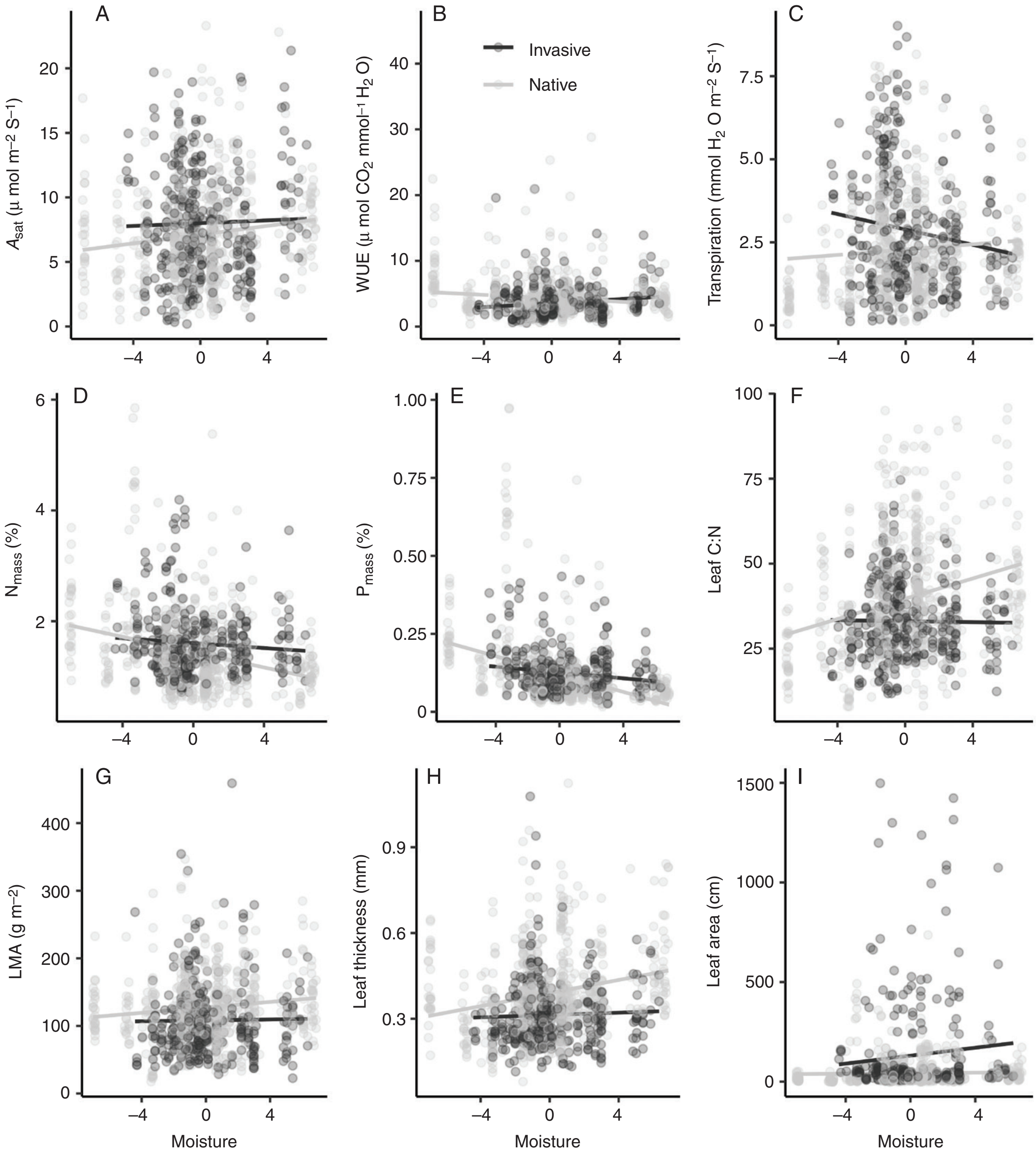

FIG. 2. Shifts in plant traits of native Hawaiian and non-native invasive species in response to moisture availability (combines soil moisture and rainfall). Each point represents one individual plant. Only traits with statistically significant origin × climate interactions are shown ($\alpha = 0.05$).

Decomposing trait variation

Plant physiological traits expressed greater intraspecific variation than leaf nutrient and morphological traits in both native and invasive species, which may reflect greater plasticity or greater population differentiation due to local adaptation in physiological traits. The key drivers of the variation were (on average, and in descending order of relative importance): species origin (native or invasive), solar energy, moisture and species identity, indicating that native and invasive species occur at different positions along the leaf economic spectrum, and that climate plays a greater role than species identity in driving variation among island plants. This is a significant finding given that species means are often used rather than individual data (Kattge *et al.*, 2011). While natives and invasives expressed similar amounts of ITV, species identity explained more trait variability in invasives than it did in natives, suggesting that species means may be more suitable for studies that focus on

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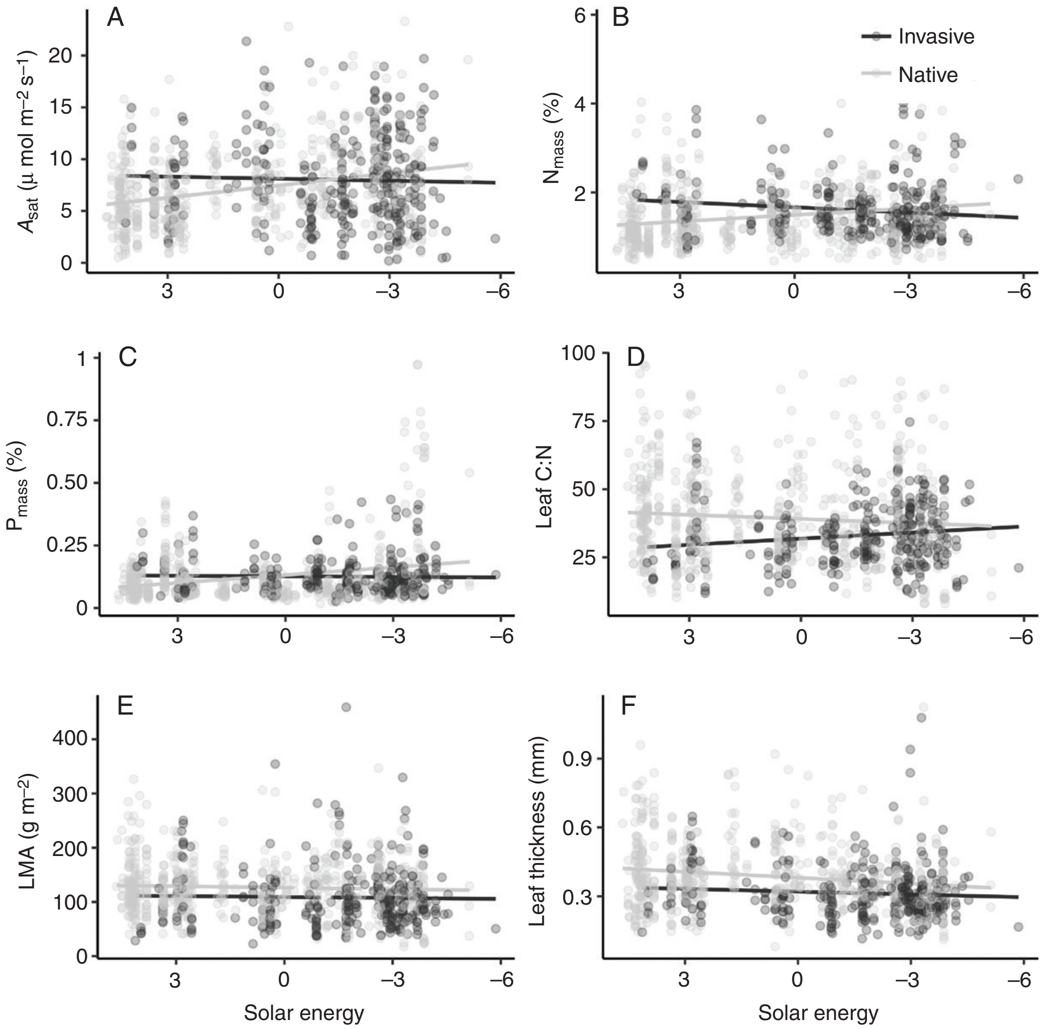

FIG. 3. Shifts in plant traits of native Hawaiian and non-native invasive species in response to solar energy (combines temperature and potential evapotranspiration). Each point represents one individual plant. Only traits with statistically significant origin × climate interactions are shown (α = 0.05).

invasives in Hawaii and less suitable for studies of native species. The higher species-level variation among invasives may reflect greater phylogenetic breadth in invasives than within the native flora or, alternatively, may reflect greater variation in morphology or the presence of more functional groups (e.g. herbs, shrubs, nitrogen-fixing trees and non-nitrogen-fixing trees) (Poorter *et al*., 2015) among invasives.

We predicted that climate would strongly influence trait variation and, indeed, moisture and solar energy contributed to 29 and 18 % of the variation on average, respectively. Averaging across all traits, available soil moisture and rainfall (captured by the moisture PCA ordinate) were stronger determinants of trait variation than temperature, evapotranspiration and elevation combined (the solar energy ordinate), suggesting that the availability of water exerts a particularly strong selective pressure and is an important source of phenotypic plasticity or population differentiation for plant traits in Hawaii. Variation in leaf nutrient content was most strongly influenced by moisture, whereas traits associated with carbon assimilation, leaf energy balance and water loss (photosynthesis, WUE, leaf area and chlorophyll content) were most strongly associated with solar energy. These results suggest that leaf nutrient content is influenced by the transpiration stream (i.e. bulk flow of water moving from roots to leaves moves nutrients up from the soil and into leaves), which is in turn driven by soil moisture (Salazar-Tortosa *et al*., 2018). These findings also indicate that light and temperature are more proximate controls of leaf energy balance and carbon assimilation in Hawaii, as temperature regulates the activity of the photosynthetic enzyme Rubisco (Bernacchi *et al*., 2001) and drives variation in leaf size (Wright *et al*., 2017). Global studies have similarly detected stronger variation in plant traits resulting from temperature and irradiance than rainfall (Moles *et al*., 2014), although it is widely recognized that these variables co-regulate plant traits (Wright *et al*., 2005). When we restricted our analysis to natives, the importance of water availability was stronger than when natives and invasives were combined, which supports previous studies that have described the sensitivity of native Hawaiian plant performance to moisture availability (Kagawa *et al*., 2009; Cavaleri *et al*., 2014). Ours is not the first study to detect significant trait variation over climate gradients in Hawaii, although the scope of our study expands considerably

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Table 1. *Summary of resource-acquisition strategies of native Hawaiian and non-native invasive species over gradients in moisture (rainfall and humidity) and available solar energy (potential evapotranspiration and air temperature)*

	Which group is more resource acquisitive? Levels of Moisture		
Trait	Low	Intermediate	High
Photosynthesis	Invasive	Invasive	Similar
WUE	Invasive	Similar	**Native**
Transpiration	Invasive	Invasive	Similar
N_{mass}	Similar	Invasive	Invasive
P_{mass}	**Native**	Similar	Invasive
Carbon:nitrogen	Similar	Invasive	Invasive
LMA	Invasive	Invasive	Invasive
Leaf thickness	Invasive	Invasive	Invasive
Leaf area	Invasive	Invasive	Invasive
	Which group is more resource acquisitive? Levels of Solar Energy		
Trait	Low	Intermediate	High
Photosynthesis	Invasive	Similar	**Native**
N_{mass}	Invasive	Similar	**Native**
P_{mass}	Invasive	Similar	**Native**
Carbon:nitrogen	Invasive	Invasive	Similar
LMA	Invasive	Invasive	Invasive
Leaf thickness	Invasive	Invasive	Invasive

The more resource-acquisitive group was determined by visually comparing the regression lines in Figs 2 and 3, which show significant origin × environment interactions for each trait separately.

Boldface indicates environmental conditions under which native species demonstrate more resource-acquisitive traits than invasives.

Resource-acquisitive traits are defined as higher photosynthesis, lower WUE, higher transpiration, higher N_{mass}, higher P_{mass}, lower carbon:nitrogen, lower LMA, lower leaf thickness and higher leaf area.

upon previous studies, which have largely been conducted on one or two islands. Tsujii *et al.* (2016) found that tissue LMA (lamina without trichomes) and leaf physical strength were best explained by temperature, while leaf area was best explained by rainfall in *Metrosideros polymorpha*, the dominant canopy species across Hawaii's forests, and Henn *et al.* (2019) detected strong shifts in leaf traits across climate gradients within Hawaii Volcanoes National Park.

Differences between natives and invasives

In a companion study, we used a multivariate analysis to investigate trait differences between native and invasive plants, and found that species origin explained only 1 % of the trait variation when analysed at the archipelago scale (A. C. Westerband *et al.*, unpubl. data). We predicted that the failure to detect differences at the archipelago scale may have been confounded by variation within species due to climate. Therefore, in the present study, we expected that differences between native and invasives would be enhanced when we captured climate-driven variation among species. While we did detect significant differences between natives and invasives in morphological and elemental composition, including mean leaf area, leaf thickness, leaf nutrient availability and chlorophyll content, the physiological traits did not vary with species origin. These findings are surprising given that the physiological traits were more variable across sites than the morphological and leaf nutrient traits. This finding may be explained by differences in the temporal scale over which morphological and physiological traits respond to changes in local climates. Physiological traits are thought to be more phenotypically plastic than morphological traits (Bradshaw, 1965), and so may be highly variable with respect to microclimate which overwhelms any potential difference between native and invasive species.

Other studies have similarly found that natives and invasives do not strictly align with resource-conservative vs. -acquisitive strategies, contrary to predictions (Tecco *et al.*, 2010; Drenovsky *et al.*, 2012; Heberling and Fridley, 2013). Using a multiscale approach, Gross *et al.* (2013) found that native and non-native species expressed similar traits at the regional scale, but these patterns were disrupted by particular species when evaluated at the community scale (e.g. a widespread native grass being taller than any non-native species at particular sites). Similarly, Funk *et al.* (2017) found that invasives were shifted towards more resource-acquisitive trait space when evaluated across five Mediterranean climate regions, but the differences were more nuanced when regions were treated individually. Both studies suggest that the scale dependency of these comparisons may be driven by variation in predominant life forms at smaller scales, e.g. grasses vs. trees and shrubs. Future studies in Hawaii should therefore integrate multiscale approaches that control for underlying variation in life forms.

We detected various climates under which natives and invasives expressed similar mean trait values (e.g. leaf nutrient content), which suggests that in many Hawaiian forests, differences in carbon economy probably play little to no role in the displacement of native by invasive plants. We even detected five (of 45) cases where native species expressed more resource-acquisitive traits than invasives, contrary to our predictions (Table 1). More commonly (30 of 45 cases), we found conditions where invasives expressed more resource-acquisitive traits, consistent with predictions and previous studies (Baruch and Goldstein, 1999; Durand and Goldstein, 2001; Cordell *et al.*, 2002; Kagawa *et al.*, 2009; Cavaleri *et al.*, 2014; Henn *et al.*, 2019). For example, in low moisture environments, we detected higher WUE in natives than in invasives, and higher transpiration in invasives, which supports previous studies that describe natives as more resource use efficient and conservative than invasives (Kagawa *et al.*, 2009). However, transpiration and WUE in natives and invasives were strikingly similar in high moisture environments, and this may result from weaker stomatal regulation in wet sites. In other words, in low moisture environments, natives reduce evaporative water loss by inducing stomatal closure, which simultaneously reduces carbon gain (isohydric), while invasives continue to transpire and photosynthesize at the same rate when moisture decreases (anisohydric). All else being equal, these differences in resource use strategies could ultimately result in greater carbon gain in invasives, leading to faster growth (Salazar-Tortosa *et al.*, 2018), but could eventually be a liability to invasives if drought increases to the point at which a more conservative stomatal regulation is advantageous. Our results may explain why at the archipelago scale, natives and invasives have overlapping traits, while at the local scale (in some environments but not

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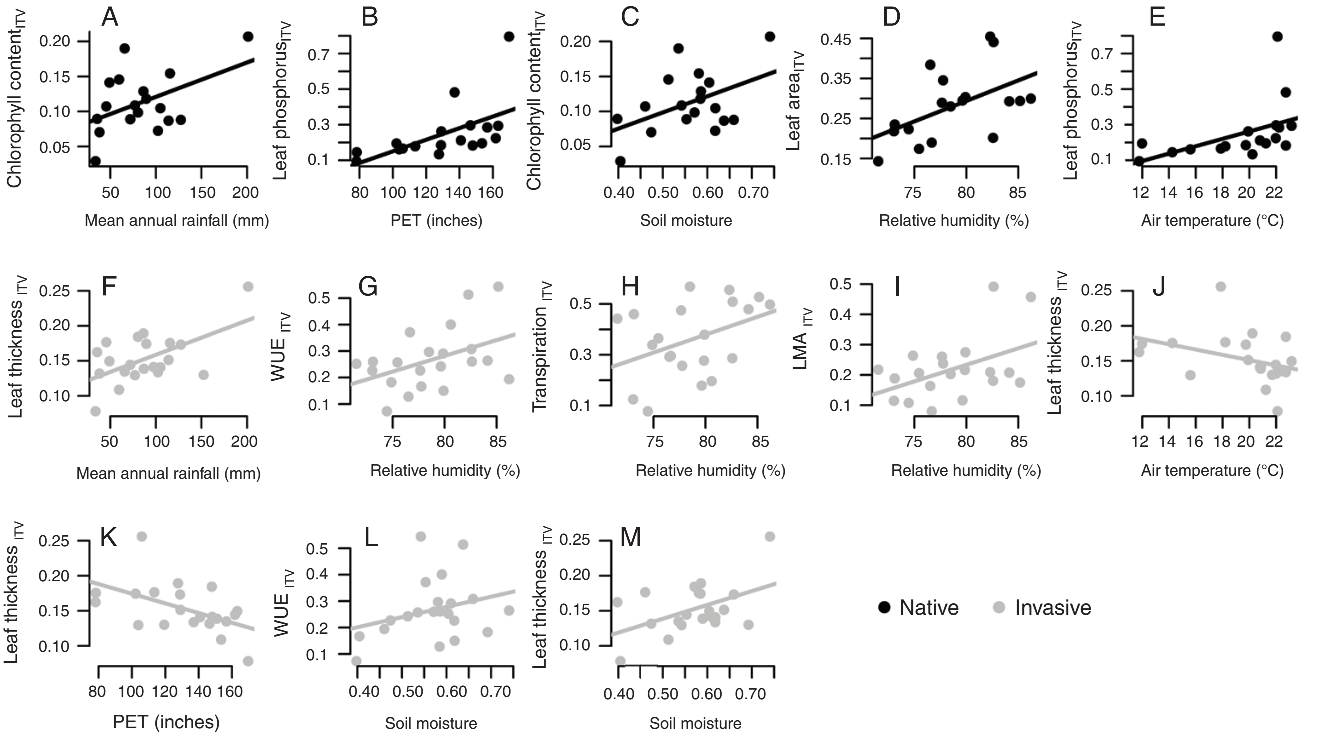

FIG. 4. Intraspecific variation in plant traits for native Hawaiian (black) and non-native invasive species (grey) in response to key abiotic gradients. Each point represents an average of all species at one study site. Only statistically significant relationships are shown (α = 0.05). PET = potential evapotranspiration.

others), there is evidence of significant trait differences. Future studies could investigate whether invasives and natives have shifted their trait spaces (e.g. from the 'fast' to the 'slow' end of the LES) in response to climate, and whether the occupied trait spaces differ across Hawaii's four largest islands. This would clarify the extent to which trait relationships are modulated or constrained by environment and intraspecific variation, which has been shown to produce deviations from the LES at regional scale (Niinemets, 2015; Asner *et al.*, 2016).

In both natives and invasives, we found evidence of less ITV within drier and hotter environments (with the exception of leaf thickness, which became more variable in hot environments), which may be indicative of trait convergence due to habitat filtering. This is consistent with the 'join-the-locals' hypothesis (Crawley *et al.*, 1996; Tecco *et al.*, 2010), which posits that non-native and native species growing in the same habitat should not differ in their traits when habitat filtering is strong. Alternatively, the 'try-harder' hypothesis argues that invasives should express more conservative traits than natives in resource-poor environments, but more acquisitive traits in resource-rich environments (e.g. highly disturbed habitats), for which we found support in terms of WUE and P_{mass} over gradients of moisture. Tecco *et al.* (2010) argue that in both cases, differences between natives and non-natives should be minimal when considered at the regional scale because differences observed at local scales (with particular environmental conditions) will be averaged together.

Interestingly, ITV was not influenced by the same sets of climate variables in natives and invasives. For example, hotter and wetter environments were associated with greater ITV in nutrient content, leaf area and chlorophyll in natives, whereas hotter environments were linked with less ITV in leaf thickness in invasives. The discrepancy may result from differences in residence times such that native Hawaiian species, which are likely to have had sufficient time for local adaptation to the environment, should have lower ITV resulting from directional selection on their traits. In contrast, invasive species may have lower ITV in more arid environments as a result of habitat filtering rather than local adaptation. To disentangle the underlying drivers of ITV would require reciprocal transplant experiments to test for local adaptation in natives and invasives.

Use of species means and future directions

Species harbour extensive variation in traits that may result from phenotypic plasticity and local adaptation (Albert *et al.*, 2010). Plasticity and local adaptation are important mechanisms that are likely to enable species' persistence under global threats, such as climate change and anthropogenically driven disturbances. On the other hand, trait variability may also explain invader success when phenotypic plasticity increases a potential invader's capacity to express advantageous traits in a broad range of environments (Bradshaw, 1965; Whitlock, 1996; Sultan, 2001), and colonize novel environments (Donohue *et al.*, 2001). While ITV has been used to clarify native–invasive dynamics at the scale of one to a few species (Funk, 2008; Drenovsky *et al.*, 2012; Sandel and Low, 2019), at the community scale it is more typical for natives and invasives to be assessed using species means. Although species means can be

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particularly informative at a global scale, it limits our understanding regarding key aspects of invasive–native dynamics (Hulme and Bernard-Verdier, 2018). For example, biotic resistance due to niche complementarity of native vs. invasive species is commonly predicted when there is trait overlap between native and invasive species (Funk *et al*., 2008). When species have extensive ITV due to large biogeographic distributions or high plasticity, trait overlap is best assessed using individual plant trait data rather than species means.

Using an archipelago-wide data set, our findings have revealed that climate drives significant variation in the direction and magnitude of trait differences between natives and invasives. Native Hawaiian plants expressed more resource-acquisitive traits than invasives in a small portion of the climate space we evaluated. However, trait differences between natives and invasives were weak or absent in many climates, indicating that other factors, such as biotic interactions or variation in below-ground traits, underlie displacement of native by invasive species in those sites. Similar to the present study, Henn *et al*. (2019) found that native and non-native species converged in their leaf thickness, specific leaf area and %C under cool, wet conditions, while non-natives were distinctly more resource acquisitive in hot and dry conditions. Thus, mitigation of invasive plant threats and conservation of highly endemic native flora need to explicitly consider climate and its role in determining the expression of traits that may confer a competitive advantage to natives over invasives. Moreover, a better understanding of how local climates will shift native–invader dynamics requires studies that measure traits over a range of environments (rather than relying on species means) while simultaneously including indices of whole-plant performance (e.g. height and reproductive output) (Hulme and Bernard-Verdier, 2018). Future studies should also investigate the extent to which changes in soil nutrient availability influence variation in plant traits, as soil is often an integral driver of plant trait variation (Ordoñez *et al*., 2009; Maire *et al*., 2015).

While the goal of this study was not to determine whether plasticity or local adaptation underlying ITV is driving invader success in Hawaii, the two processes are not mutually exclusive and could both play important roles in the ITV detected in this study. Godoy *et al*. (2011) detected evidence of plasticity and local adaptation in response to light availability for an invasive perennial herb, which probably increases its capacity to colonize the understorey of South American temperate evergreen forests. Similarly, both adaptation and phenotypic plasticity enabled an invasive herb to acclimate to localized variation in soil moisture and light, which explained its success within a tropical island in China at regional scale (Si *et al*., 2014). These results suggest that more effective conservation of native species may be achieved by considering how performance varies across a mosaic of habitats that occur over the scale of a few kilometres.

Conclusions

The present study clarifies the extent to which climate drives trait variation in native Hawaiian species, and the extent to which invasives vary in their trait expression within the same environments. We detected similar ITV in natives and invasives, but these were constrained by distinct climate variables. Furthermore, in a sub-set of the climate space, natives expressed more resource-acquisitive traits than invasives, which is contrary to many previous studies. Significant climate-driven variation in leaf traits explains why we failed to detect significant differences between natives and invasives at the archipelago-scale in a companion study (A. C. Westerband *et al*., unpubl. data), and why localized studies may detect significant differences between natives and invasives while others do not. That we detected evidence of increasing ITV but also increasingly resource-acquisitive traits in natives as temperature and potential evapotranspiration increased suggests that native Hawaiian plants have the capacity to outcompete non-native invaders under particular environmental conditions. These findings also offer new insights regarding the potential role of climate in determining invader success at regional vs. local scales.

SUPPLEMENTARY DATA

Supplementary data are available online at https://academic.oup.com/aob and consist of the following. Table S1: replication per species per site. Table S2: replication per species across all sites. Table S3: correlations between abiotic variables and principal component axes. Table S4: calculation of intraspecific trait variation. Table S5: comparisons of mixed models with or without climate. Table S6: significance of fixed effects from models shown in Table S5. Table S7: significance of interactions between species origin and climate. Table S8: effect of climate on intraspecific trait variation. Figure S1: principal components analysis of climate variables for natives and invasives. Figure S2: magnitude of intraspecific variation in each trait.

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